



## KALN

Aluminum Nitride

## KEY PROPERTIES

- High thermal conductivity
- Electrical insulator
- Good mechanical properties
- Thermal expansion similar to semi-conductors

## APPLICATIONS

- Heat sinks & heat spreaders
- Electronic insulating components
- Substrates for microelectronic devices

## MAIN PROPERTIES

| Property                     | Value | Unit                              |
|------------------------------|-------|-----------------------------------|
| Density                      | 3.30  | g/cm <sup>3</sup>                 |
| Flexural Strength            | 330   | MPa                               |
| Porosity                     | 0     | %                                 |
| Thermal Conductivity (100°C) | >165  | W/m·K                             |
| CTE (20-500°C)               | 4.5   | x10 <sup>-6</sup> K <sup>-1</sup> |
| Volume Resistivity           | 1E13  | Ω·cm                              |
| Dielectric strength          | >15   | kV/mm                             |

\* All properties measured at 20°C unless otherwise stated.

\*\* Measured in a wall thickness of 2 mm

## OPERATING CONDITIONS

|                          |      |    |
|--------------------------|------|----|
| Max working temp (air)   | 1350 | °C |
| Max working temp (inert) | 1900 | °C |
| Recomended temp          | 1500 | °C |
| Continuous temp          | 1400 | °C |

## ELECTRICAL BEHAVIOR

|                     |     |
|---------------------|-----|
| Dielectric strength | >15 |
|---------------------|-----|

## MATERIAL INFORMATION

|               |               |
|---------------|---------------|
| Material type | AlN (pure)    |
| Compability   | Yes           |
| Structure     | Dense ceramic |
| Purity        | High          |

The information provided is for guidance only and does not constitute a guarantee.

Nanoker Research S.L.

Polígono de Olloniego, Parcela 21A, Nave 4-8, 33660, Oviedo (Asturias), Spain

(+34) 985 20 76 13 | [info@nanoker.com](mailto:info@nanoker.com)